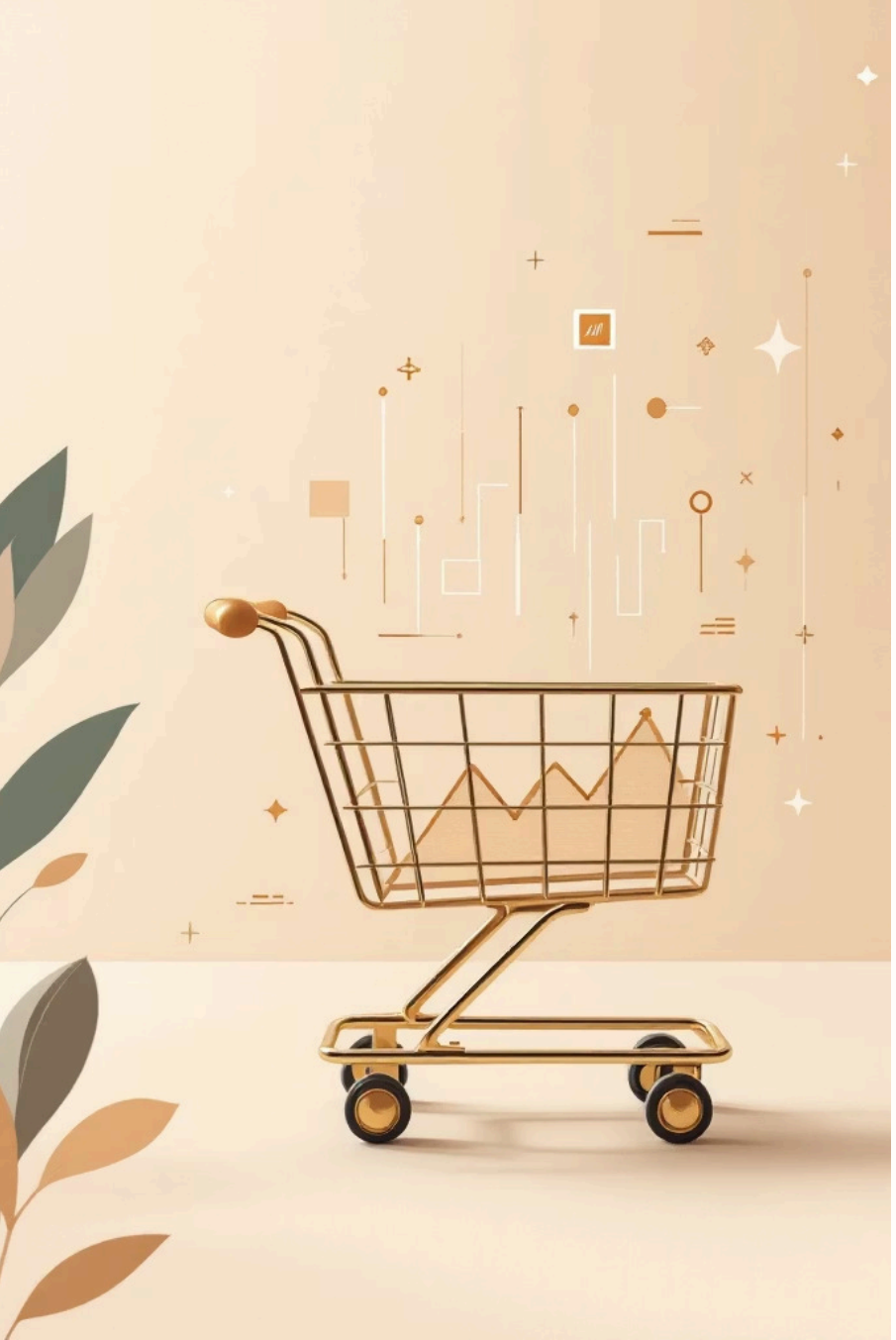
Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Made with GAMMA



Project Overview

Dataset Size

3,900 purchases across 18 columns of customer data

Goal

Uncover spending patterns, preferences, and subscription behavior

Features

Demographics, purchase details, and shopping behavior metrics

Made with GAMMA

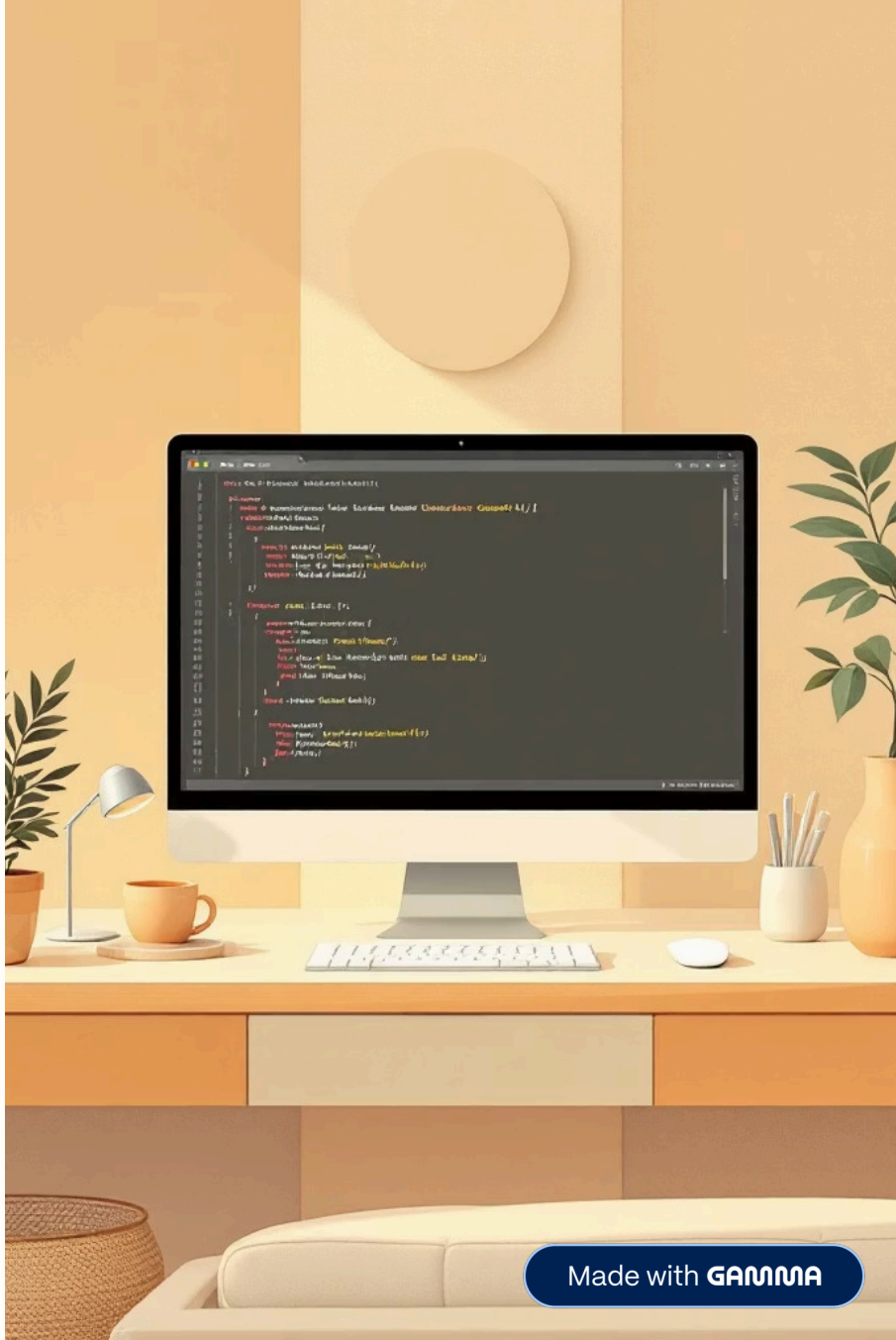
Data Preparation Process

Python Analysis

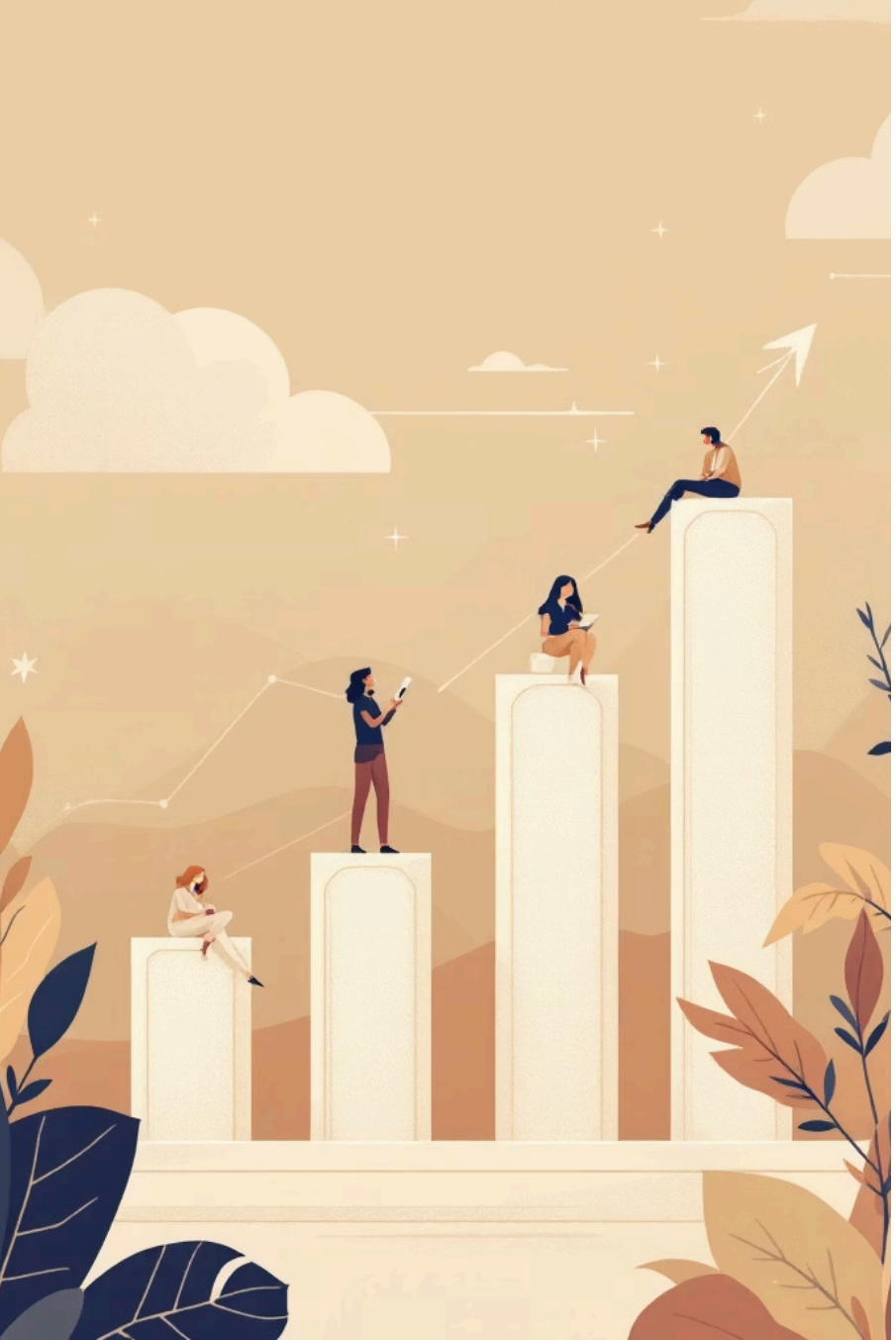
- Data loading with pandas
- Initial exploration and summary statistics
- Missing data imputation using median ratings
- Column standardization to snake case

Feature Engineering

- Created age groups by binning
- Calculated purchase frequency in days
- Dropped redundant promo code column
- Loaded to PostgreSQL for SQL analysis



Made with GAMMA



SQL Analysis: Revenue Insights

$157,890

Male Revenue

Significantly higher than female customers

$75,191

Female Revenue

Lower but steady purchasing pattern

839

Discount Users

High-spending customers using discounts

Made with GAMMA

Product Performance

Gloves

Top-rated at 3.86 average rating

Sandals

Second highest at 3.84 rating

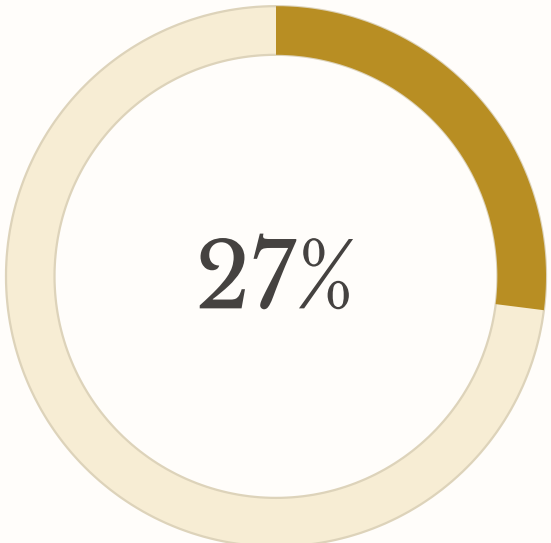
Boots

Third place with 3.82 rating

Shipping comparison showed Express shipping customers spent $60.48 average vs. $58.46 for Standard.

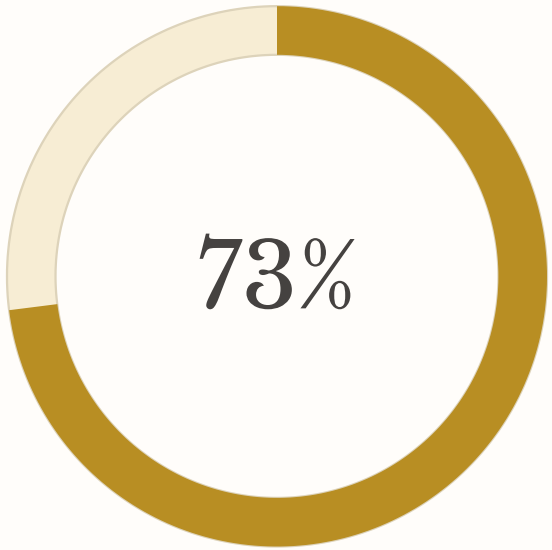
Made with GAMMA

Subscription Analysis



Subscribers

1,053 customers with subscription



Non-Subscribers

2,847 customers without subscription

Average Spend

Subscribers: $59.49

Non-Subscribers: $59.87

Total Revenue

Subscribers: $62,645

Non-Subscribers: $170,436

Made with GAMMA

Discount Strategy Insights

1

Hats

50% of purchases discounted

2

Sneakers

49.66% discount rate

3

Coat

49.07% discounted purchases

4

Sweater

48.17% discount usage

5

Pants

47.37% discounted transactions

High discount rates on top products suggest effective sales strategy but may impact margins.

Made with GAMMA

Customer Segmentation

Loyal

3,116 customers with consistent purchase history

Returning

701 customers making repeat purchases

New

83 customers with single purchase

Made with GAMMA

Age Group Revenue Analysis

27%

Young Adult

$62,143 revenue

25%

Middle-aged

$59,197 revenue

24%

Adult

$55,978 revenue

24%

Senior

$55,763 revenue

Young adults drive highest revenue, followed by middle-aged, adult, and senior segments.

Made with GAMMA

Strategic Recommendations

01

Boost Subscriptions

Promote exclusive benefits to increase subscriber base from current 27%

02

Enhance Loyalty Programs

Reward repeat buyers to move them into loyal customer segment

03

Optimize Discount Policy

Balance sales boosts with margin control on high-discount products

04

Strategic Product Positioning

Highlight top-rated products (gloves, sandals, boots) in marketing campaigns

05

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

Made with GAMMA